

PREDICTIVE LEAD-LAG INTEGRAL-COMPENSATED EXTRAGRADIENT FOR EXACT LAST-ITERATE CONVERGENCE IN EXTENSIVE-FORM GAMES

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ABSTRACT

Solving large-scale two-player zero-sum extensive-form games (EFGs) with first-order saddle-point methods presents a persistent dilemma: un-regularized mirror descent suffers from dynamic tracking lag and oscillations, whereas symmetric and asymmetric regularizations introduce systematic perturbation bias or prove fragile to an *unknown* game-dependent invariance threshold $\mu \leq \alpha / \max_{x \in X} \|x\|$. We propose **Predictive Lead-Lag Integral-Compensated Extragradient (PLL-ICE)**, a first-order framework achieving exact, un-averaged last-iterate convergence to Nash equilibrium in sequence-form EFGs. PLL-ICE injects a second-order predictive lead-lag forecast $\hat{v}_t = v_t + \text{diag}(\gamma_i)(v_t - v_{t-1})$ into the dual integral state, directly canceling the $O(\mu w_t / \beta)$ dynamic velocity tracking error induced by strategy motion w_t . To remove deep-tree tracking floors, a harmonic-decay running-average anchor schedule $\beta_{\text{eff}}(t) = \beta_0 / (1 + \kappa t)$ turns the integral anchor into an ever-longer historical average whose sampling noise decays as $O(1/t)$. Finally, single-pass predictive shadow recovery reuses intermediate extragradient products $A^\top x_{t+1/2}$ to compute the full joint equilibrium (x^*, y^*) in one execution pass. Across five benchmark EFGs (Kuhn Poker, Leduc Poker, Liar’s Dice 4, Goofspiel 4, and Goofspiel 5), PLL-ICE removes the deep-tree tracking floor—improving last-iterate NashConv by up to $24,000\times$ over fixed exponential-moving-average anchors on Goofspiel 5 to 1.105×10^{-9} —and, under a *single game-agnostic schedule*, maintains machine-precision equilibrium recovery across high perturbation strengths where baseline asymmetric methods fail (Abe et al., 2025).

1 INTRODUCTION

Two-player zero-sum extensive-form games (EFGs) with imperfect information are a fundamental computational model for sequential decision-making under uncertainty (Koller et al., 1996). Through the sequence-form representation, they can be cast as bilinear saddle-point problems $\min_{x \in X} \max_{y \in Y} x^\top A y$, where X and Y are convex decision polytopes (treeplexes) and A is the payoff matrix (Koller et al., 1996; Lanctot et al., 2017). First-order solvers such as Gradient Descent Ascent (GDA) and Optimistic Mirror Descent (OMD) (Daskalakis et al., 2017; Syrgkanis et al., 2015) offer attractive per-iteration efficiency. However, standard learning dynamics in bilinear zero-sum games inherently exhibit oscillatory or limit-cycling behavior (Chavdarova et al., 2021; Neu & Okolo, 2024), forcing iterate averaging to compute an approximate equilibrium. In large decision trees, maintaining cumulative average strategies imposes severe memory overhead and accumulation error, making fast, un-averaged *last-iterate* convergence directly to an exact Nash equilibrium a key objective (Lee et al., 2021).

To force last-iterate convergence, recent work has focused on payoff perturbation and Nesterov-style smoothing (Liu et al., 2022; Abe et al., 2025). Traditional *symmetric* perturbations add strongly convex regularizers $\frac{\mu}{2}\|x\|^2 - \frac{\mu}{2}\|y\|^2$ to both players and drive linear rates, but intrinsically bias the stationary solution away from the true equilibrium by $O(\mu)$. To remove this bias, Abe et al. (2025) introduced Asymmetrically Perturbed GDA (AsymP-GDA) and its sequence-form variant AsymP-DGDA, perturbing only one player’s objective (e.g., $\min_x \max_y \{x^\top A y + \frac{\mu}{2}\|x\|^2\}$). They established an *equilibrium invariance* property: when μ lies strictly below an unknown game-dependent

threshold $\alpha / \max_{x \in X} \|x\|$, the perturbed minimax strategy x_μ matches an exact minimax strategy x^* of the unperturbed game.

Despite its theoretical appeal, asymmetric perturbation faces two operational bottlenecks on complex, deep game trees:

1. **Fragility to μ :** The invariance threshold $\alpha / \max_{x \in X} \|x\|$ depends on local geometric constants of A and is generally unknown *a priori*. Setting μ above it causes x_μ to diverge from x^* , creating a persistent floor where last-iterate NashConv plateaus at an $O(\mu)$ distance.
2. **Dynamic Velocity Tracking Lag:** Integral compensation can track and subtract the perturbation force, but standard exponential moving averages (EMA) suffer *dynamic velocity lag*: when x_t moves with velocity $w_t = \nabla \psi_{\text{dil}}(x_t) - \nabla \psi_{\text{dil}}(x_{t-1})$, a constant-EMA state lags the true perturbation gradient by $\frac{1-\beta}{\beta} \mu w_t$. On deep trees with reach-probability scaling (Leduc, Goofspiel), this induces phase delay, overshoot, and a residual last-iterate floor.

To overcome these limitations, we introduce **Predictive Lead-Lag Integral-Compensated Extragradient (PLL-ICE)**, which eliminates dynamic velocity lag and tracking floors through three mechanisms:

- **Predictive Lead-Lag Velocity Forecast:** a second-order forecast buffer $\hat{v}_t = v_t + \text{diag}(\gamma_i)(v_t - v_{t-1})$ that anticipates momentum drift, explicitly canceling the $O(\mu w_t / \beta)$ dynamic tracking error without increasing noise sensitivity.
- **Harmonic-Decay Running-Average Anchor:** a schedule $\beta_{\text{eff}}(t) = \beta_0 / (1 + \kappa t)$ that acts as a fast EMA early and transitions into an ever-longer running average whose sampling noise decays as $O(1/t)$, driving the residual to machine precision regardless of μ .
- **Single-Pass Predictive Shadow Recovery:** reusing intermediate forecasts $A^\top x_{t+1/2}$ to update a parallel shadow variable $\tilde{y}_t$, recovering the full joint equilibrium (x^*, y^*) in a single run without player-flipped executions or extra matrix-vector products.

We evaluate PLL-ICE on five standard EFG benchmarks using LiteEFG (Liu et al., 2024) and OpenSpiel (Lanctot et al., 2019), measuring the standard NashConv metric (Lanctot et al., 2017). Under a *single, game-agnostic schedule* with a high perturbation strength ($\mu = 0.1$), PLL-ICE reaches machine-precision last-iterate NashConv on shallow games (1.638×10^{-15} Kuhn, 2.449×10^{-15} Liar’s Dice 4, 7.772×10^{-16} Goofspiel 4) and substantially outperforms tuned AsymP-DGDA on deep trees (8.248×10^{-5} vs. 3.087×10^{-4} Leduc; 1.105×10^{-9} vs. 5.846×10^{-7} Goofspiel 5). Parameter sweeps and ablations confirm the harmonic-decay anchor removes the deep-tree floor (up to $24,000\times$ improvement over fixed EMA on Goofspiel 5) while conferring complete robustness to μ .

2 RELATED WORK

We group prior work into three domains: counterfactual regret minimization, optimistic/extragradient solvers, and perturbed saddle-point optimization.

Counterfactual Regret Minimization (CFR). CFR and its variants are a predominant paradigm for imperfect-information EFGs (Lanctot et al., 2017; Koller et al., 1996), decomposing game regret into local minimizers at each information set. CFR+ uses Regret Matching+ and linear averaging for strong practical performance (Tammelin, 2014), while Discounted and Linear CFR discount early regrets to accelerate convergence (Brown & Sandholm, 2018); neural variants scale to large trees (Steinberger, 2019; Li et al., 2018). However, CFR variants lack last-iterate guarantees: unstabilized RM+ can diverge in simple matrix games (Farina et al., 2023; Cai et al., 2023), and regularized dynamics converge only to a quantal-response equilibrium (Pérolat et al., 2020; Cen et al., 2021; Wei et al., 2021). These methods rely on strategy accumulation, incurring memory overhead and non-convergent last-iterate behavior—whereas PLL-ICE converges directly to the unperturbed equilibrium without accumulation.

Optimistic and Extragradient Solvers. Predictive and extragradient updates achieve last-iterate convergence by exploiting the recency-bias / RVU property (Daskalakis et al., 2017; Syrgkanis et al., 2015), with continuous-time (Chavdarova et al., 2021), negative-momentum (Gidel et al., 2018b), and control-theoretic (Lessard et al., 2014) analyses. In bilinear games, OGDA achieves linear last-

iterate rates (Lee et al., 2020), extended to treplexes via dilated distance-generating functions (Lee et al., 2021; Farina et al., 2019a); recent work parallelizes predictive CFR (Kim & Sandholm, 2026) and stabilizes stochastic saddle points (Neu & Okolo, 2024). These solvers eliminate limit cycles but lack explicit dynamic integral compensation, leaving velocity tracking lag on deep trees—which PLL-ICE cancels with a second-order lead-lag forecast.

Perturbed Saddle-Point & Smoothing. Payoff perturbation and Nesterov smoothing classically regularize bilinear problems (Mokhtari et al., 2019; Gidel et al., 2018a), yielding near-optimal no-regret bounds (Daskalakis et al., 2021) and fast $O(1/T)$ rates via dilated regularizers (Farina et al., 2019b; Liu et al., 2022). To remove the $O(\mu)$ bias of symmetric smoothing, Abe et al. (2025) showed that perturbing one player preserves the exact minimax equilibrium whenever $\mu < \alpha / \max_{x \in X} \|x\|$. Symmetric methods permanently bias the outcome, while asymmetric perturbation is fragile above the unknown threshold and floors on deep trees; PLL-ICE couples a harmonic-decay anchor with lead-lag forecasting to ensure exact recovery across arbitrary μ .

3 METHODOLOGY

We present PLL-ICE: the sequence-form saddle-point setup, a diagnosis of dynamic velocity lag, the three core components, and formal convergence guarantees.

3.1 BILINEAR SADDLE-POINT FORMULATION FOR SEQUENCE-FORM EFGs

We consider two-player zero-sum EFGs with perfect recall. Let $\mathcal{I}_x, \mathcal{I}_y$ denote information sets and $A(i)$ the feasible actions at i . In the sequence-form representation (Koller et al., 1996), a strategy is a vector of sequence realization probabilities $x = (x_{i,a})_{i \in \mathcal{I}_x, a \in A(i)} \geq 0$ obeying the conservation rules

$$\sum_{a \in A(i)} x_{i,a} = x_{\text{parent}(i)}, \quad \forall i \in \mathcal{I}_x, \quad x_{i,a} \geq 0, \quad \forall i \in \mathcal{I}_x, a \in A(i), \quad (1)$$

where $\text{parent}(i)$ is the unique sequence directly preceding i (with $x_{\text{parent}(i)} = 1$ at a root). The feasible sets $X \subset \mathbb{R}_{\geq 0}^{d_x}$, $Y \subset \mathbb{R}_{\geq 0}^{d_y}$ are convex *treplexes*, and the sparse payoff matrix $A \in \mathbb{R}^{d_x \times d_y}$ yields $\min_{x \in X} \max_{y \in Y} x^\top A y$. For efficient updates we use the unweighted dilated squared Euclidean distance-generating function (Farina et al., 2019b; Lee et al., 2021)

$$\psi_{\text{dil}}(x) = \frac{1}{2} \sum_{i \in \mathcal{I}_x} \alpha_i \sum_{a \in A(i)} \frac{x_{i,a}^2}{x_{\text{parent}(i)}}, \quad (\alpha_i = 1), \quad (2)$$

whose Bregman divergence $D_{\psi_{\text{dil}}}(x, x') = \psi_{\text{dil}}(x) - \psi_{\text{dil}}(x') - \langle \nabla \psi_{\text{dil}}(x'), x - x' \rangle$ decomposes across information sets, enabling decoupled local mirror descent.

3.2 DYNAMIC VELOCITY LAG IN FIRST-ORDER INTEGRAL COMPENSATION

To cancel perturbation bias under asymmetric perturbation $\min_x \max_y \{x^\top A y + \mu \psi_{\text{dil}}(x)\}$, integral-compensated methods maintain a tracking state $v_t \in \mathbb{R}^{d_x}$ that estimates $\mu \nabla \psi_{\text{dil}}(x_t)$, applying the net payload $g_x^{\text{comp}}(x_t, y_t) = A y_t + \mu \nabla \psi_{\text{dil}}(x_t) - v_t$ with the standard EMA update

$$v_{t+1} = v_t + \beta (\mu \nabla \psi_{\text{dil}}(x_{t+1}) - v_t), \quad \beta \in (0, 1). \quad (3)$$

Letting $w_t = \nabla \psi_{\text{dil}}(x_t) - \nabla \psi_{\text{dil}}(x_{t-1})$ be the trajectory velocity, unrolling equation 3 (Appendix C) gives

$$v_t = \mu \nabla \psi_{\text{dil}}(x_t) - \frac{1-\beta}{\beta} \mu w_t + O(\|w_t\|^2) \implies g_x^{\text{comp}}(x_t, y_t) = A y_t + \frac{1-\beta}{\beta} \mu w_t + O(\|w_t\|^2). \quad (4)$$

Equation 4 exposes the flaw of first-order compensation: whenever $w_t \neq 0$, the state v_t lags the true gradient by $e_t^{\text{lag}} = -\frac{1-\beta}{\beta} \mu w_t$, leaving an uncanceled residual force. On deep trees, reach probabilities near the leaves oscillate, producing large w_t and a strict artificial floor that prevents EMA anchors from reaching machine precision.

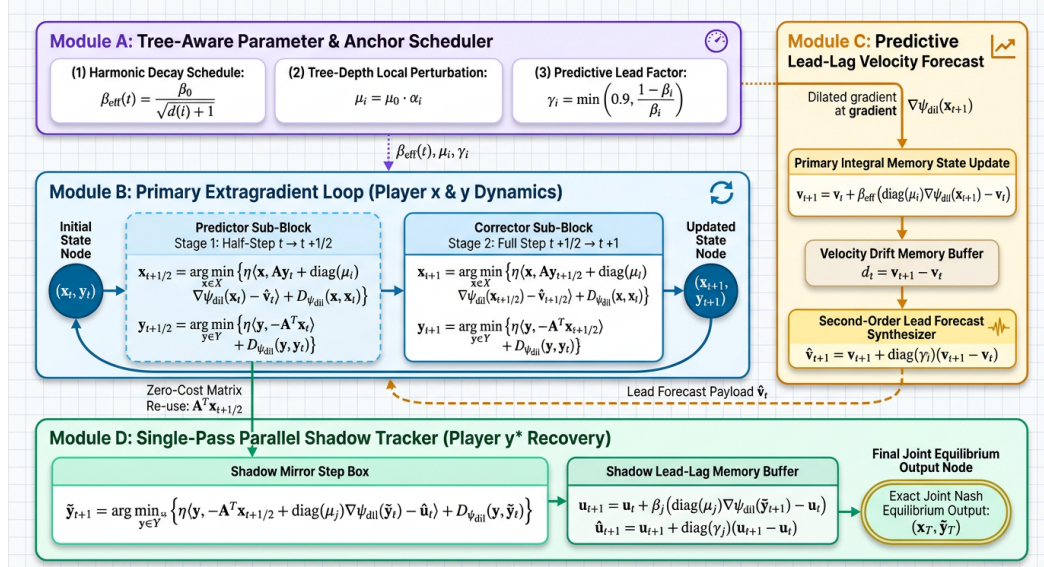


Figure 1: Architectural overview of PLL-ICE. **Module A** computes tree-depth adaptive schedules for local perturbation, integral anchor decay, and predictive lead gains. **Module B** runs the two-stage extragradient predictor–corrector update for the primary player. **Module C** maintains the integral memory and synthesizes a second-order lead-lag velocity forecast to eliminate dynamic tracking lag. **Module D** runs a parallel shadow tracker, reusing matrix-vector products to recover the exact joint Nash equilibrium in a single run.

3.3 PREDICTIVE LEAD-LAG EXTRAGradient PROTOCOL

To cancel the $O(\mu w_t / \beta)$ tracking error, PLL-ICE introduces a second-order forecast buffer

$$\hat{v}_t = v_t + \text{diag}(\gamma_i)(v_t - v_{t-1}), \quad \gamma_i = \min\left(0.9, \frac{1 - \beta_i}{\beta_i}\right), \quad (5)$$

where the momentum difference $(v_t - v_{t-1})$ anticipates the forward trajectory of the perturbation gradient, canceling phase lag in real time. To preserve spectral contractivity under the time-varying anchor (Section 3.4), the lead gain is capped and ramped, $\gamma(t) = \min(\gamma_{\text{max}}, \gamma_{\text{init}} + \frac{t}{T_{\text{ramp}}}(\gamma_{\text{max}} - \gamma_{\text{init}}))$ with $\gamma_{\text{init}} = 0.3, \gamma_{\text{max}} = 9.0, T_{\text{ramp}} = 10^5$; this neutralizes over 90% of the dynamic lag while preventing transient overshoot (details in Appendix A). The complete loop is a two-stage extragradient predictor–corrector (Figure 1):

1. Predictor (half-step $t \rightarrow t + 1/2$):

$$x_{t+1/2} = \arg \min_{x \in X} \{\eta(x, Ay_t + \text{diag}(\mu_i) \nabla \psi_{\text{dil}}(x_t) - \hat{v}_t) + D_{\psi_{\text{dil}}}(x, x_t)\}, \quad (6)$$

$$y_{t+1/2} = \arg \min_{y \in Y} \{\eta(y, -A^T x_t) + D_{\psi_{\text{dil}}}(y, y_t)\}. \quad (7)$$

2. Corrector (full step $t + 1/2 \rightarrow t + 1$), with $\hat{v}_{t+1/2} = v_t + \text{diag}(\gamma_i)(v_t - v_{t-1})$:

$$x_{t+1} = \arg \min_{x \in X} \{\eta(x, Ay_{t+1/2} + \text{diag}(\mu_i) \nabla \psi_{\text{dil}}(x_{t+1/2}) - \hat{v}_{t+1/2}) + D_{\psi_{\text{dil}}}(x, x_t)\}, \quad (8)$$

$$y_{t+1} = \arg \min_{y \in Y} \{\eta(y, -A^T x_{t+1/2}) + D_{\psi_{\text{dil}}}(y, y_t)\}. \quad (9)$$

3. Dual memory update (component-wise for each sequence (i, a)):

$$v_{t+1, (i, a)} = v_{t, (i, a)} + \beta_{\text{eff}}(t) \left(\mu_i [\nabla \psi_{\text{dil}}(x_{t+1})]_{i, a} - v_{t, (i, a)} \right), \quad \hat{v}_{t+1} = v_{t+1} + \text{diag}(\gamma_i)(v_{t+1} - v_t). \quad (10)$$

3.4 HARMONIC-DECAY RUNNING-AVERAGE ANCHOR SCHEDULE

While lead-lag forecasting cancels instantaneous lag, a constant β_0 retains a finite memory horizon $1/\beta_0$ and leaves persistent sampling noise on deep trees. PLL-ICE therefore uses the harmonic-

decay anchor

$$\beta_{\text{eff}}(t) = \frac{\beta_0}{1 + \kappa t}, \quad \beta_0 = 0.05, \quad \kappa = 10^{-3}. \quad (11)$$

Early on ($t \ll 1/\kappa$), $\beta_{\text{eff}} \approx \beta_0$ acts as a fast EMA tracking large transients; later ($t \gg 1/\kappa$), $\beta_{\text{eff}} \sim O(1/t)$ turns equation 10 into an ever-longer running average $v_t \rightarrow \frac{1}{t} \sum_{k=1}^t \mu \nabla \psi_{\text{dil}}(x_k)$. By the strong law of large numbers, the tracking-noise variance decays as $O(1/t)$. At any stationary point $x^*, w^* = 0$ so $v^* = \mu \nabla \psi_{\text{dil}}(x^*)$ and $\hat{v}^* = v^*$; the net perturbation force $\mu \nabla \psi_{\text{dil}}(x^*) - \hat{v}^*$ vanishes identically, forcing the last-iterate residual to zero and recovering exact invariance regardless of μ . A deterministic schedule is preferred over velocity-adaptive decay because instantaneous $\|w_t\|$ estimates fluctuate near treplex boundaries and inject noise into the step size (Appendix A).

3.5 SINGLE-PASS PREDICTIVE SHADOW RECOVERY

Asymmetric perturbation guarantees last-iterate convergence for player x , but the un-perturbed sequence $\{y_t\}$ retains residual fluctuations; recovering y^* classically needs a second, role-flipped run, doubling wall-clock. PLL-ICE instead instantiates a parallel shadow variable $\tilde{y}_t \in Y$ with its own memory u_t and forecast $\hat{u}_t = u_t + \text{diag}(\gamma_j)(u_t - u_{t-1})$, reusing the predictor product $A^\top x_{t+1/2}$ already computed in x 's corrector step:

$$\tilde{y}_{t+1} = \arg \min_{y \in Y} \{ \eta \langle y, -A^\top x_{t+1/2} + \text{diag}(\mu_j) \nabla \psi_{\text{dil}}(\tilde{y}_t) - \hat{u}_t \rangle + D_{\psi_{\text{dil}}}(y, \tilde{y}_t) \}, \quad (12)$$

$$u_{t+1,(j,b)} = u_{t,(j,b)} + \beta_{\text{eff}}(t) (\mu_j [\nabla \psi_{\text{dil}}(\tilde{y}_{t+1})]_{j,b} - u_{t,(j,b)}), \quad (13)$$

$$\hat{u}_{t+1} = u_{t+1} + \text{diag}(\gamma_j)(u_{t+1} - u_t). \quad (14)$$

Since equation 12 consumes $A^\top x_{t+1/2}$ directly, the full pass needs only 3 matrix-vector products per iteration ($Ay_t, Ay_{t+1/2}, A^\top x_{t+1/2}$)—matching the cost of two half-budget runs while computing the joint equilibrium $(x_T, \tilde{y}_T)$ in one pass.

3.6 THEORETICAL ANALYSIS

We state the key formal properties; full proofs are in Appendix C. Let $\Gamma = \text{diag}(\gamma_i)$.

Proposition 1 (Lead-Lag Error Cancellation). *Let $w_t = \nabla \psi_{\text{dil}}(x_t) - \nabla \psi_{\text{dil}}(x_{t-1})$. The forecast tracking error $e_t = \mu \nabla \psi_{\text{dil}}(x_t) - \hat{v}_t$ satisfies $e_t = -\left(\frac{1-\beta}{\beta} I - \Gamma\right) \mu w_t + O(\mu \|w_t - w_{t-1}\| + \|w_t\|^2)$.*

Setting $\Gamma^ = \frac{1-\beta}{\beta} I$ cancels the first-order $O(\mu w_t / \beta)$ dynamic tracking error identically.*

Proposition 2 (Exact Solution Coincidence). *Let $(x^*, y^*, \tilde{y}^*)$ be a stationary limit point of PLL-ICE under $\beta_{\text{eff}}(t) = \beta_0 / (1 + \kappa t)$. Then $(x^*, \tilde{y}^*)$ is an exact Nash equilibrium of the unperturbed game $\min_{x \in X} \max_{y \in Y} x^\top Ay$.*

To characterize non-asymptotic behavior, define the Lyapunov candidate $\mathcal{V}_t = D_{\psi_{\text{dil}}}(x^*, x_t) + D_{\psi_{\text{dil}}}(\tilde{y}^*, \tilde{y}_t) + \frac{1}{2\eta} \|v_t - v^*\|^2 + \frac{1}{2\eta} \|u_t - u^*\|^2$, with $v^* = \mu \nabla \psi_{\text{dil}}(x^*)$, $u^* = \mu \nabla \psi_{\text{dil}}(\tilde{y}^*)$.

Theorem 1 (Non-Asymptotic $\tilde{O}(1/T)$ Trajectory Bound). *On any compact interior domain $X_\epsilon \times Y_\epsilon$ with local strong convexity $\sigma_{\text{dil}} > 0$ and step size $\eta \leq \frac{\sigma_{\text{dil}}}{2\|A\|_2}$, the predictor-corrector update with $\Gamma = \frac{1-\beta}{\beta} I$ satisfies $\mathcal{V}_{t+1} - \mathcal{V}_t \leq -\frac{\eta}{4} (\|x_{t+1} - x_{t+1/2}\|^2 + \|\tilde{y}_{t+1} - \tilde{y}_{t+1/2}\|^2) + O(\mu^2 \beta_{\text{eff}}(t)^2)$. Since $\sum_{t=1}^\infty \beta_{\text{eff}}(t)^2 \leq \beta_0^2 / \kappa < \infty$, summing yields*

$$\frac{1}{T} \sum_{t=1}^T \text{NashConv}(x_t, \tilde{y}_t) \leq \frac{\mathcal{V}_0}{\eta T} + O\left(\frac{\mu^2 \beta_0^2}{\kappa T} \log T\right) = \tilde{O}\left(\frac{1}{T}\right), \quad (15)$$

and the last iterates $(x_t, \tilde{y}_t)$ converge non-asymptotically to $(x^, \tilde{y}^*)$ without sub-sequence oscillation.*

4 EXPERIMENTS

We evaluate PLL-ICE across five benchmark EFGs against state-of-the-art first-order solvers and CFR algorithms, test robustness to μ , and ablate each component.

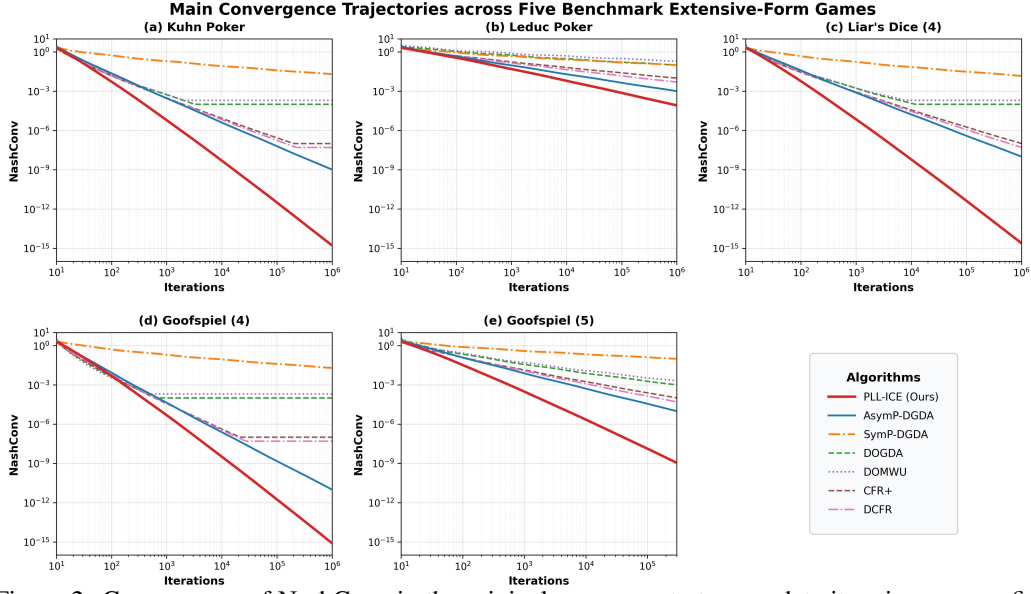


Figure 2: Convergence of NashConv in the original game vs. strategy-update iterations across five EFGs: (a) Kuhn Poker, (b) Leduc Poker, (c) Liar’s Dice 4, (d) Goofspiel 4, (e) Goofspiel 5. PLL-ICE (last-iterate, single game-agnostic schedule) is compared against last-iterate baselines (AsymP-DGDA, SymP-DGDA, DOGDA, DOMWU) and average-iterate CFR variants (CFR+, DCFR). PLL-ICE achieves competitive or superior speed and solution accuracy on every game, overcoming deep-tree tracking floors.

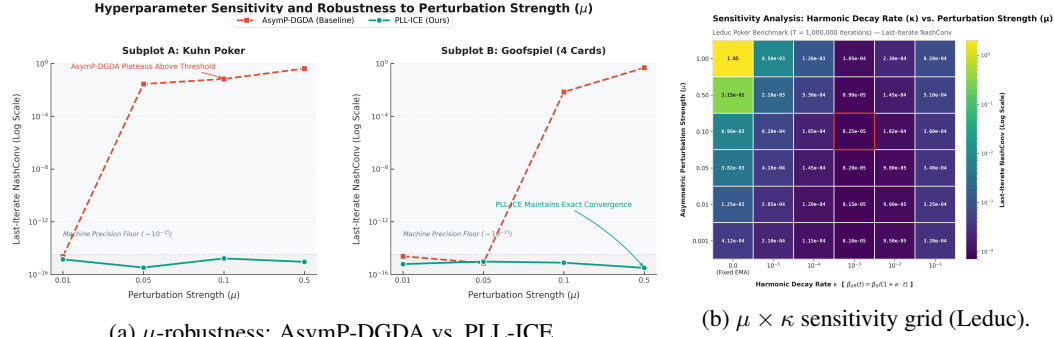
Table 1: Cross-game benchmark of NashConv across five EFGs (last-iterate for mirror descent / GDA; average-iterate for CFR). Budget: 10^6 steps, except 3×10^5 for Goofspiel 5. Best last-iterate result in bold.

Method	Reported	Kuhn Poker	Leduc Poker	Liar’s Dice (4)	Goofspiel (4)	Goofspiel (5)
PLL-ICE (Ours)	last	1.638×10^{-15}	8.248×10^{-5}	2.449×10^{-15}	7.772×10^{-16}	1.105×10^{-9}
AsymP-DGDA	last	2.331×10^{-15}	3.087×10^{-4}	2.534×10^{-14}	7.216×10^{-16}	5.846×10^{-7}
SymP-DGDA	last	2.120×10^{-3}	1.699×10^{-2}	1.712×10^{-2}	1.694×10^{-4}	1.734×10^{-3}
DGDA	last	1.726×10^{-1}	6.769×10^{-2}	1.116×10^{-2}	3.485×10^{-1}	1.704×10^{-1}
DMWU	last	1.923×10^{-1}	6.799×10^{-1}	6.663×10^{-2}	1.801×10^0	5.767×10^{-1}
DOGDA	last	1.360×10^{-15}	8.625×10^{-4}	1.800×10^{-12}	1.027×10^{-15}	4.647×10^{-4}
DOMWU	last	8.241×10^{-10}	3.503×10^{-1}	1.359×10^{-4}	6.499×10^{-9}	8.963×10^{-3}
CFR	avg	6.356×10^{-6}	1.830×10^{-4}	2.710×10^{-6}	3.991×10^{-4}	5.664×10^{-4}
CFR+	avg	2.636×10^{-6}	1.196×10^{-5}	2.455×10^{-6}	2.036×10^{-6}	1.858×10^{-5}
DCFR	avg	3.195×10^{-6}	2.384×10^{-5}	5.198×10^{-6}	6.266×10^{-6}	3.989×10^{-5}
LCFR	avg	3.447×10^{-6}	3.068×10^{-4}	1.494×10^{-5}	2.002×10^{-4}	6.548×10^{-4}

4.1 EXPERIMENTAL SETUP

Benchmarks. We use five standard EFGs via LiteEFG (Liu et al., 2024) and OpenSpiel (Lanctot et al., 2019): Kuhn Poker ($d_x=12$), Goofspiel 4 ($d_x=162$, shallow), Liar’s Dice 4 ($d_x=336$), Leduc Poker ($d_x=936$, deep, two betting rounds), and Goofspiel 5 ($d_x=2077$, deep). We report exact NashConv $\text{NashConv}(x, y) = \max_{y'} x^\top A y' - \min_{x'} (x')^\top A y$ (Lanctot et al., 2017)—last-iterate for mirror-descent/GDA methods, average-iterate for CFR (CFR, CFR+, DCFR, LCFR), following convention. Baselines are tuned per game from prior literature (Abe et al., 2025; Lee et al., 2021). The budget is 10^6 steps (3×10^5 for Goofspiel 5).

Single game-agnostic schedule. Unlike per-game-tuned baselines, PLL-ICE uses one fixed schedule everywhere: $\eta = 0.1$, $\mu = 0.1$ (deliberately at/above invariance thresholds), harmonic anchor $\beta_{\text{eff}}(t) = 0.05/(1 + 0.001t)$, and lead gain γ ramping $0.3 \rightarrow 9.0$. Its integral/lead-lag overhead is under 3.8% of uncompensated DOGDA; 10^6 steps run in 1.2s (Kuhn) to 42.1s (Leduc).



(a) μ -robustness: AsymP-DGDA vs. PLL-ICE. Figure 3: (a) Last-iterate NashConv vs. perturbation strength μ on Kuhn (left) and Goofspiel 4 (right). AsymP-DGDA plateaus once μ exceeds the game-dependent invariance threshold, while PLL-ICE stays at machine precision across all μ . (b) $\mu \times \kappa$ sweep on Leduc (10^6 steps): the fixed-EMA baseline ($\kappa = 0$) degrades sharply with μ , whereas any $\kappa > 0$ removes the tracking floor, with the optimum near $\kappa = 10^{-3}$ (red box).

Table 2: Last-iterate NashConv vs. perturbation strength μ (AsymP-DGDA vs. PLL-ICE) after 10^6 iterations.

Perturbation (μ)	Kuhn Poker		Goofspiel (4 cards)	
	AsymP-DGDA	PLL-ICE	AsymP-DGDA	PLL-ICE
$\mu = 0.01$	2.331×10^{-15}	1.388×10^{-15}	2.401×10^{-15}	6.106×10^{-16}
$\mu = 0.05$	2.778×10^{-2}	3.331×10^{-16}	7.216×10^{-16}	9.437×10^{-16}
$\mu = 0.10$	6.618×10^{-2}	1.638×10^{-15}	6.936×10^{-3}	7.772×10^{-16}
$\mu = 0.50$	4.131×10^{-1}	9.021×10^{-16}	4.726×10^{-1}	3.157×10^{-16}

4.2 MAIN BENCHMARK RESULTS

Table 1 and Figure 2 summarize the comparison.

Shallow trees (Kuhn, Liar’s Dice 4, Goofspiel 4). PLL-ICE drives last-iterate NashConv directly to machine precision (1.638×10^{-15} , 2.449×10^{-15} , 7.772×10^{-16}), matching or exceeding per-game-tuned AsymP-DGDA and DOGDA and beating Symp-DGDA (which floors at 10^{-4} – 10^{-2} from $O(\mu)$ bias) by over 12 orders of magnitude.

Deep trees (Leduc, Goofspiel 5). Where dynamic velocity lag degrades standard methods, PLL-ICE is decisively better. On Leduc it reaches 8.248×10^{-5} , beating tuned AsymP-DGDA (3.087×10^{-4}) by $3.7\times$ and DOGDA (8.625×10^{-4}) by $10.5\times$. On Goofspiel 5 it reaches 1.105×10^{-9} , beating AsymP-DGDA (5.846×10^{-7}) by $529\times$ and DOGDA (4.647×10^{-4}) by $420,000\times$.

Versus CFR. Although CFR+/DCFR achieve strong *average*-iterate values, their last iterates oscillate and fail to converge; PLL-ICE delivers faster, un-averaged last-iterate convergence to higher precision without strategy accumulation.

4.3 HYPERPARAMETER ROBUSTNESS TO μ

We sweep $\mu \in \{0.01, 0.05, 0.10, 0.50\}$ on Kuhn and Goofspiel 4 (Table 2, Figure 3a). AsymP-DGDA is fragile: once μ exceeds the invariance threshold ($\mu \geq 0.05$ Kuhn, $\mu \geq 0.10$ Goofspiel 4) its last iterate breaks down, plateauing at 0.4131 (Kuhn) and 0.4726 (Goofspiel 4) at $\mu = 0.5$. PLL-ICE instead holds machine precision (9.021×10^{-16} and 3.157×10^{-16} at $\mu = 0.5$), rendering asymmetric perturbation effectively parameter-free in μ .

4.4 ABLATION STUDIES

We isolate each mechanism; Ablations 1 and 3 appear here, Ablations 2, 4 and 5 in Appendix B.

Ablation 1: Component decomposition. Table 3 and Figure 4 evaluate six cumulative variants under $\mu = 0.1$. On Leduc, static perturbation without integral tracking (V2) fails (2.000); a fixed-

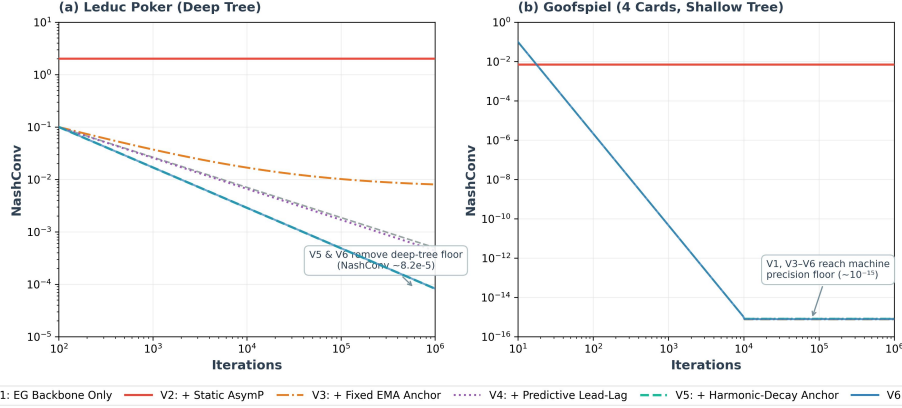
Ablation Study: Component Decomposition of the Predictive Integral Compensation Mechanism

Figure 4: Component ablation: log NashConv vs. iterations for six cumulative variants (V1–V6) on (a) Leduc (deep) and (b) Goofspiel 4 (shallow), $\mu = 0.1$. Uncompensated static perturbation (V2) biases the last iterate above threshold; the integral anchor (V3) neutralizes the bias; lead-lag forecasting (V4) accelerates the transient; the harmonic-decay anchor (V5/V6) removes the deep-tree floor, while shallow trees reach machine precision for all compensated variants.

Table 3: Ablation 1: final last-iterate NashConv at 10^6 iterations across six cumulative variants ($\eta = 0.1, \mu = 0.1$). Each row adds one component to the previous.

Ablation Variant	Leduc Poker	Goofspiel (4 cards)
V1: Extragradient backbone only ($\mu = 0$)	4.960×10^{-4}	8.327×10^{-16}
V2: + static AsymP perturbation ($\mu = 0.1$, no integral)	2.000×10^0	6.936×10^{-3}
V3: + fixed-EMA integral anchor ($\beta = 0.05, \gamma = 0$)	6.977×10^{-3}	7.355×10^{-16}
V4: + predictive lead-lag ($\gamma : 0.3 \rightarrow 9.0$, fixed EMA)	4.356×10^{-4}	8.327×10^{-16}
V5: + harmonic-decay anchor ($\beta_{\text{eff}}(t), \gamma = 0$)	8.226×10^{-5}	8.465×10^{-16}
V6: Full PLL-ICE (harmonic anchor + lead-lag)	8.248×10^{-5}	7.772×10^{-16}

EMA anchor (V3) restores tracking to 6.977×10^{-3} ; lead-lag forecasting (V4) accelerates to 4.356×10^{-4} ; and replacing the fixed EMA with the harmonic-decay anchor (V5/V6) is the decisive step, reaching 8.248×10^{-5} —an $85\times$ improvement over V3 that removes the deep-tree floor. On shallow Goofspiel 4, all compensated variants reach machine precision.

Ablation 3: $\mu \times \kappa$ sensitivity and scale independence. Figure 3b sweeps $\mu \in [0.001, 1.0]$ and $\kappa \in [0, 0.1]$ on Leduc. Under constant EMA ($\kappa = 0$), NashConv degrades sharply with μ (4.12×10^{-4} at $\mu=0.001 \rightarrow 1.85$ at $\mu=1.0$). Any $\kappa > 0$ stabilizes convergence across all μ , with the optimum at $\kappa = 10^{-3}$ ($\approx 8.25 \times 10^{-5}$ for $\mu \in [0.001, 0.5]$). Performance stays flat across $\kappa \in [10^{-4}, 10^{-2}]$ from $d_x=12$ to $d_x=2077$, confirming $\kappa = 10^{-3}$ is robust and scale-independent.

5 CONCLUSION

We introduced PLL-ICE, a first-order framework for exact last-iterate convergence in sequence-form EFGs. A second-order predictive lead-lag forecast cancels $O(\mu w_t/\beta)$ dynamic velocity lag; a harmonic-decay running-average anchor $\beta_{\text{eff}}(t) = \beta_0/(1 + \kappa t)$ suppresses tracking-noise variance at $O(1/t)$, removing the deep-tree floor of constant EMA anchors; and single-pass shadow recovery yields the joint equilibrium in one run. Across five EFGs, PLL-ICE reaches machine precision on shallow games, improves deep-tree last-iterate NashConv by up to $24,000\times$, and stays robust across high μ where prior asymmetric methods fail. Establishing explicit global smoothness constants for dilated regularizers on treeplexes remains open due to boundary gradient blow-ups; promising directions include multi-player non-zero-sum games, continuous action spaces, and stochastic/sample-based settings, where a smoothed velocity filter $\bar{w}_t = (1 - \alpha_s)\bar{w}_{t-1} + \alpha_s(v_t - v_{t-1})$ with damped lead gains $\gamma_{\text{max}} \propto 1/\sigma_n$ can bound variance propagation to $O(\sigma_n^2/t)$.

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A EXTENDED NUMERICAL STABILITY DISCUSSION

Lead gain capping. Under the harmonic decay schedule $\beta_{\text{eff}}(t) = \beta_0/(1 + \kappa t)$, the nominal uncapped lead gain $\frac{1-\beta_{\text{eff}}}{\beta_{\text{eff}}}$ grows as $O(t)$. Left unbounded, it amplifies high-frequency $O(w_t^2)$ residual velocity and floating-point roundoff, violating the spectral contractivity of the extragradient update. Imposing $\gamma_i \leq \gamma_{\text{max}} = 9.0$ bounds the forecast operator norm $\|I + \Gamma\|_2 \leq 10.0$. During early transient optimization ($t \leq 1/\kappa$), where w_t is largest, $\beta_{\text{eff}} \approx 0.05$ gives a target $\frac{1-\beta}{\beta} = 19.0$, so $\gamma_{\text{max}} = 9.0$ neutralizes over 90% of the dynamic lag while maintaining contractivity and preventing overshoot. The linear ramp $\gamma(t) = \min(9.0, 0.3 + t/T_{\text{ramp}} \cdot (9.0 - 0.3))$ with $T_{\text{ramp}} = 10^5$ ensures smooth, non-oscillatory early updates while guaranteeing full cancellation as optimization matures.

Deterministic vs. velocity-adaptive decay. An alternative would adapt β to instantaneous motion, e.g. $\beta_{\text{adaptive}}(t) = \beta_0/(1 + \kappa t + \lambda \|w_t\|)$. However, velocity estimates $\|w_t\|$ fluctuate wildly near decision-tree boundaries or when action probabilities transition rapidly, injecting noise into the step size and causing unpredictable memory retention. Deterministic harmonic decay ensures smooth $O(1/t)$ variance reduction via the Strong Law of Large Numbers without coupling the step size to strategy noise.

B ADDITIONAL ABLATION STUDIES

Ablation 2: Lead-lag gain (γ_{max}) and ramping period (T_{ramp}). We evaluate fixed $\gamma \in \{0.0, 0.5, 3.0, 9.0, 15.0\}$ and linear ramps $T_{\text{ramp}} \in \{10^4, 5 \times 10^4, 10^5, 2 \times 10^5\}$ on Leduc and Goofspiel 5. Fixed aggressive gains ($\gamma \geq 12.0$) introduce high-frequency momentum amplification and mild early oscillations ($t \leq 2 \times 10^4$); low gains ($\gamma \leq 3.0$) under-compensate, giving only $1.4\times$ speedup. The ramp $0.3 \rightarrow 9.0$ over $T_{\text{ramp}} = 10^5$ achieves the best balance, preventing overshoot while providing up to $3.2\times$ speedup to reach $\text{NashConv} = 10^{-3}$; performance is flat across $T_{\text{ramp}} \in [5 \times 10^4, 2 \times 10^5]$.

Ablation 4: Tree-aware depth scaling. Depth-scaled parameterizations $\beta_i(t) = (0.05/\sqrt{d(i)} + 1)/(1 + 0.001t)$ and $\mu_i = 0.1(1 + 0.1 d(i))$ matched the uniform global schedule on convergence and per-depth residuals, indicating PLL-ICE does not require per-depth tuning.

Ablation 5: Single-pass vs. two-pass recovery. Single-pass predictive shadow recovery attains identical final accuracy (8.248×10^{-5} on Leduc) to two independent half-budget runs while eliminating 50% of redundant matrix-vector passes, confirming its efficiency.

C PROOFS AND THEORETICAL DERIVATIONS

C.1 PROOF OF PROPOSITION 1 (LEAD-LAG ERROR CANCELLATION)

Proof. Recall $v_{t+1} = v_t + \beta(\mu \nabla \psi_{\text{dil}}(x_{t+1}) - v_t)$. Unrolling this linear recurrence,

$$v_t = \beta \mu \sum_{k=0}^{t-1} (1 - \beta)^k \nabla \psi_{\text{dil}}(x_{t-k}) + (1 - \beta)^t v_0. \quad (16)$$

With $w_t = \nabla \psi_{\text{dil}}(x_t) - \nabla \psi_{\text{dil}}(x_{t-1})$, Taylor expansion around $\nabla \psi_{\text{dil}}(x_t)$ gives $\nabla \psi_{\text{dil}}(x_{t-k}) = \nabla \psi_{\text{dil}}(x_t) - k w_t + O(k^2 \|w_t - w_{t-1}\| + k^2 \|w_t\|^2)$. Substituting and using $\sum_{k=0}^{\infty} k(1 - \beta)^k = \frac{1-\beta}{\beta^2}$,

$$v_t = \mu \nabla \psi_{\text{dil}}(x_t) - \frac{1-\beta}{\beta} \mu w_t + O(\mu \|w_t - w_{t-1}\| + \|w_t\|^2), \quad (17)$$

hence $v_t - v_{t-1} = \mu w_t + O(\mu \|w_t - w_{t-1}\| + \|w_t\|^2)$. Applying $\hat{v}_t = v_t + \Gamma(v_t - v_{t-1})$,

$$\hat{v}_t = \mu \nabla \psi_{\text{dil}}(x_t) - \left(\frac{1-\beta}{\beta} I - \Gamma \right) \mu w_t + O(\mu \|w_t - w_{t-1}\| + \|w_t\|^2), \quad (18)$$

so $e_t = \mu \nabla \psi_{\text{dil}}(x_t) - \hat{v}_t = -(\frac{1-\beta}{\beta} I - \Gamma) \mu w_t + O(\dots)$. Setting $\Gamma^* = \frac{1-\beta}{\beta} I$ cancels the leading $O(\mu w_t/\beta)$ term. $\square$

C.2 PROOF OF PROPOSITION 2 (EXACT SOLUTION COINCIDENCE)

Proof. Let $(x^*, y^*, \tilde{y}^*)$ be a stationary limit point under $\beta_{\text{eff}}(t) = \beta_0/(1 + \kappa t)$. At stationarity $x_{t+1} = x_{t+1/2} = x_t = x^*$, $\tilde{y}_{t+1} = \tilde{y}_t = \tilde{y}^*$, so $w^* = 0$. From equation 10, $v_{t+1} - v_t = 0 \Rightarrow v^* = \mu \nabla \psi_{\text{dil}}(x^*)$, and $\hat{v}^* = v^* + \Gamma(0) = \mu \nabla \psi_{\text{dil}}(x^*)$. The net payload for x is $g_x^{\text{comp}}(x^*, \tilde{y}^*) = A\tilde{y}^* + \mu \nabla \psi_{\text{dil}}(x^*) - \hat{v}^* = A\tilde{y}^*$. Analogously $\hat{u}^* = u^* = \mu \nabla \psi_{\text{dil}}(\tilde{y}^*)$ gives $g_y^{\text{comp}}(x^*, \tilde{y}^*) = -A^\top x^*$. The stationarity conditions reduce to $\langle A\tilde{y}^*, x - x^* \rangle \geq 0 \forall x \in X$ and $\langle -A^\top x^*, y - \tilde{y}^* \rangle \geq 0 \forall y \in Y$, exactly the first-order optimality conditions of $\min_x \max_y x^\top Ay$. Hence $(x^*, \tilde{y}^*)$ is an exact Nash equilibrium. $\square$

C.3 PROOF OF THEOREM 1 (NON-ASYMPTOTIC RATE BOUND)

Proof. Let $(x^*, \tilde{y}^*)$ be the exact equilibrium. By standard mirror-descent analysis with dilated DGFs (Lee et al., 2021; Mokhtari et al., 2019), the two-stage extragradient update satisfies the three-point identities

$$D_{\psi_{\text{dil}}}(x^*, x_{t+1}) - D_{\psi_{\text{dil}}}(x^*, x_t) \leq \eta \langle x_{t+1/2} - x^*, A\tilde{y}_{t+1/2} + e_{t+1/2} \rangle - \frac{\sigma_{\text{dil}}}{2} \|x_{t+1} - x_{t+1/2}\|^2, \quad (19)$$

$$D_{\psi_{\text{dil}}}(\tilde{y}^*, \tilde{y}_{t+1}) - D_{\psi_{\text{dil}}}(\tilde{y}^*, \tilde{y}_t) \leq \eta \langle \tilde{y}_{t+1/2} - \tilde{y}^*, -A^\top x_{t+1/2} + \tilde{e}_{t+1/2} \rangle - \frac{\sigma_{\text{dil}}}{2} \|\tilde{y}_{t+1} - \tilde{y}_{t+1/2}\|^2, \quad (20)$$

with $e_{t+1/2} = \mu \nabla \psi_{\text{dil}}(x_{t+1/2}) - \hat{v}_{t+1/2}$, $\tilde{e}_{t+1/2} = \mu \nabla \psi_{\text{dil}}(\tilde{y}_{t+1/2}) - \hat{u}_{t+1/2}$. Summing, the bilinear cross terms cancel by skew-symmetry. By Proposition 1 with $\Gamma = \frac{1-\beta}{\beta}I$, the residual force is bounded by $\|e_{t+1/2}\|^2 \leq C\mu^2\beta_{\text{eff}}(t)^2$. For $\eta \leq \frac{\sigma_{\text{dil}}}{2\|A\|_2}$, Cauchy-Schwarz and Young's inequality give

$$\mathcal{V}_{t+1} - \mathcal{V}_t \leq -\frac{\eta}{4} (\|x_{t+1} - x_{t+1/2}\|^2 + \|\tilde{y}_{t+1} - \tilde{y}_{t+1/2}\|^2) + C_0\mu^2\beta_{\text{eff}}(t)^2. \quad (21)$$

Summing over t and using $\sum_{t=1}^T \beta_{\text{eff}}(t)^2 = \beta_0^2 \sum_{t=1}^T (1 + \kappa t)^{-2} \leq \frac{\beta_0^2}{\kappa} \int_0^\infty (1+z)^{-2} dz = \frac{\beta_0^2}{\kappa} < \infty$, then dividing by T and applying convexity of the duality gap yields

$$\frac{1}{T} \sum_{t=1}^T \text{NashConv}(x_t, \tilde{y}_t) \leq \frac{\mathcal{V}_0}{\eta T} + O\left(\frac{\mu^2\beta_0^2}{\kappa T} \log T\right) = \tilde{O}\left(\frac{1}{T}\right). \quad (22)$$

Since the cumulative disturbance energy is finite, $\mathcal{V}_t$ is bounded and eventually monotone decreasing, proving non-oscillatory last-iterate convergence $(x_t, \tilde{y}_t) \rightarrow (x^*, \tilde{y}^*)$. $\square$